(4) If D and E are the midpoints of the sides AB and AC of a triangle ABC, prove that $\overrightarrow{BE}+\overrightarrow{DC}=\frac{3}{2}\overrightarrow{BC}$.

(5) Prove that the line segment joining the midpoints of two sides of a triangle is parallel to the third side whose length is half of the length of the third side.

(6) Prove that the line segments joining the midpoints of the adjacent sides of a quadrilateral form a parallelogram.

(7) If $\vec{a}$ and $\vec{b}$ represent a side and a diagonal of a parallelogram, find the other sides and the other diagonal.

(8) If $\overrightarrow{PO}+\overrightarrow{OQ}=\overrightarrow{QO}+\overrightarrow{OR}$, prove that the points P, Q, R are collinear.

(9) If D is the midpoint of the side BC of a triangle ABC, prove that $\overrightarrow{AB}+\overrightarrow{AC}=2\overrightarrow{AD}$.

(10) If G is the centroid of a triangle ABC, prove that $\overrightarrow{GA}+\overrightarrow{GB}+\overrightarrow{GC}=\vec{0}$.

(11) Let A, B, and C be the vertices of a triangle. Let D,E, and F be the midpoints of the sides BC, CA, and AB respectively. Show that $\overrightarrow{AD}+\overrightarrow{BE}+\overrightarrow{CF}=\vec{0}$.

(12) If $ABCD$ is a quadrilateral and E and F are the midpoints of AC and BD respectively, then prove that $\overrightarrow{AB}+\overrightarrow{AD}+\overrightarrow{CB}+\overrightarrow{CD}=4\overrightarrow{EF}$.

8.6 Resolution of Vectors

Resolution of a vector can be done for any finite dimension. But we will discuss only in two and three dimensions. Let us start with two dimension.

8.6.1 Resolution of a vector in two dimension

Theorem 8.5

Let $\hat{i}$ and $\hat{j}$ be the unit vectors along the positive x-axis and the y-axis having initial point at the origin O. Now $\overrightarrow{OP}$ is the position vector of any point P in the plane. Then $\overrightarrow{OP}$ can be uniquely written as

$\overrightarrow{OP} = x\hat{i}+y\hat{j}$ for some real numbers x and y. Further $|\overrightarrow{OP}|=\sqrt{x^2+y^2}$.

Proof

Let (x, y) be the coordinates of the point P. Let L and M be the foots of the perpendiculars drawn from P to the x and y axes. Then $\overrightarrow{OP}=\overrightarrow{OL}+\overrightarrow{LP}=\overrightarrow{OL}+\overrightarrow{OM}$.

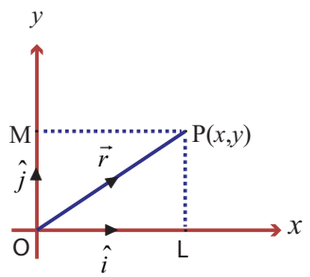


Fig. 8.29

Since $\hat{i}$ and $\hat{j}$ are unit vectors, we have $\overrightarrow{OL}=x\hat{i}$ and $\overrightarrow{OM}=y\hat{j}$.

Thus $\overrightarrow{OP} = x\hat{i}+y\hat{j}$

If $\overrightarrow{OP}=\vec{r}$ then $\vec{r}=x\hat{i}+y\hat{j}$.

To prove the uniqueness, let $x_1\hat{i}+y_1\hat{j}$ and $x_2\hat{i}+y_2\hat{j}$ be two representations of the same point P. Then

$$x_1\hat{i}+y_1\hat{j} = x_2\hat{i}+y_2\hat{j}.$$

This implies that $(x_1-x_2)\hat{i}-(y_2-y_1)\hat{j}=\vec{0}\Rightarrow x_1-x_2=0, y_2-y_1=0.$

In other words $x_1=x_2$ and $y_1=y_2$ and hence the uniqueness follows.

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